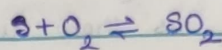


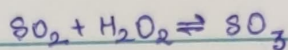
Course Content

a) Some chemical reactions relevant to process industries

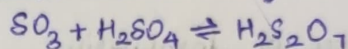
b) S and H_2SO_4 (contact process)



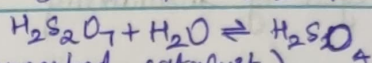
c) Polymer fundamentals



d) Petroleum refinery industries



e) Catalyst ~~separation~~ preparation (supported catalyst)



+ active metal/oxide

For sulphur:

There are some rules of support catalyst

1) Frasch Process

2) H_2S oxidation and reduction

3) Finnish process

} processes to extract Sulphur

Polymers:

a) Monomeric Unit

(i) $=$ (Double bond) (Functionality = 2)

(ii) $\equiv$ (Triple bond) (Functionality = 4)

(iii) Functional group (reactive)

↳ $COOH$ (Carboxylic group)

NH_2 (Amide)

OH (Hydroxide)

↓
Alcohol group

Three different processes:

(i) High Pressure process

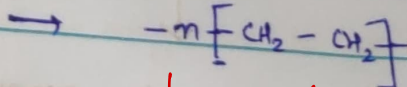
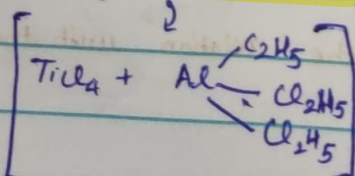
(ii) Intermediate "

(iii) Low " "

↳ $70^\circ C$

6-10 atm

Ziegler-Natta Catalyst



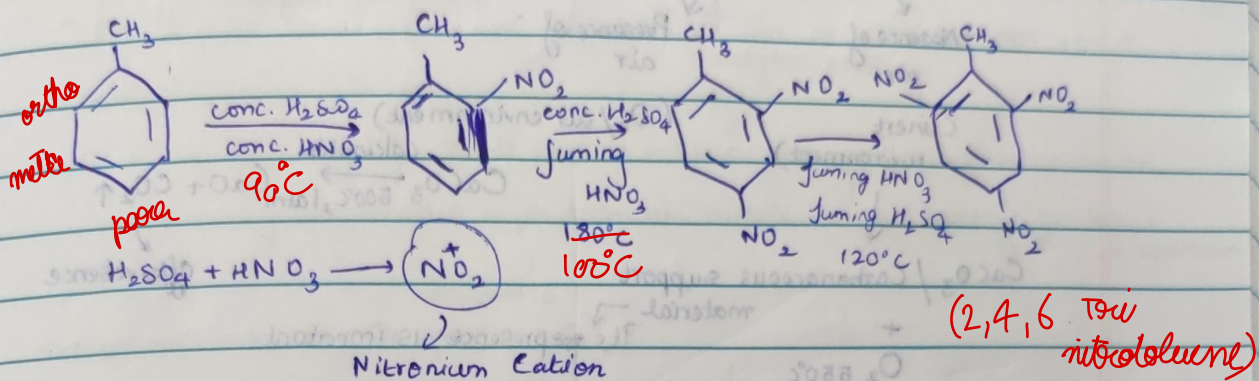
↳ no. of repeat units

① Saturated Compound

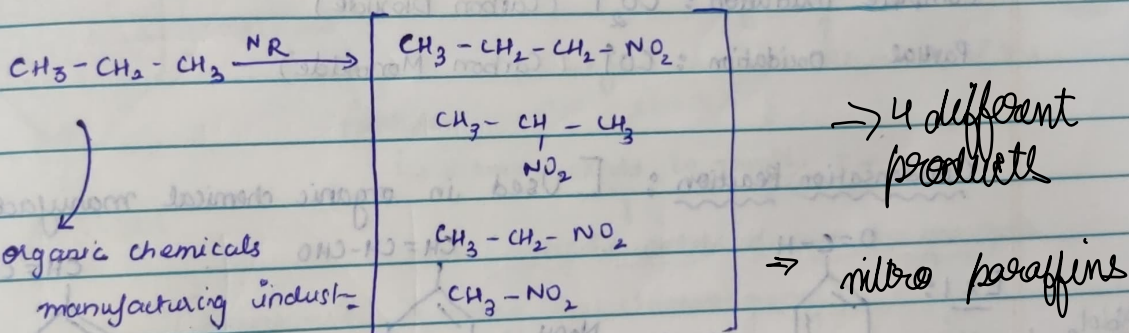
② Unsaturated compound [cyclic structure without double and triple bond]

③ Naphthenic compound

④ Aromatic compound



Nitration Rxn's

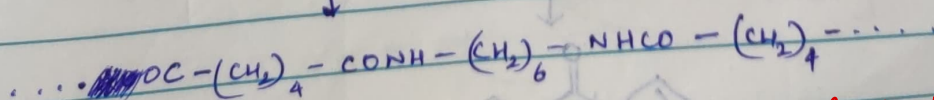
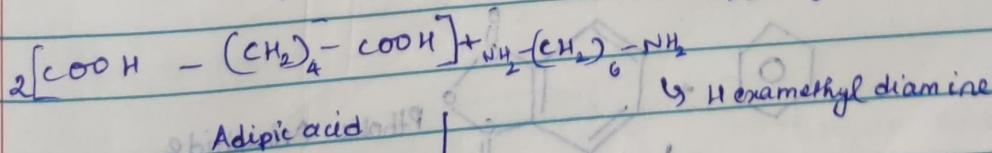


Polymerization Rxn:

① Addition polymerization

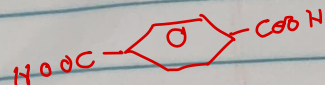
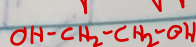
② Condensation Polymerization $\rightarrow$ formation of by products

↓
removal of small molecules
↓
 $\text{CO}_2, \text{NH}_3, \text{H}_2\text{O} \dots \text{etc}$



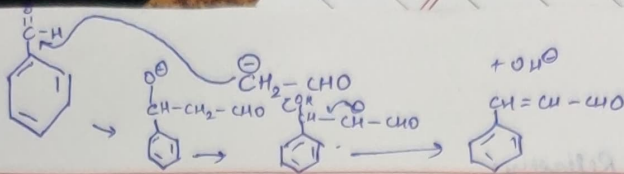
Nylon 6,6

Ethylene glycol + Terephthalic acid



↳ Alkyd Resin

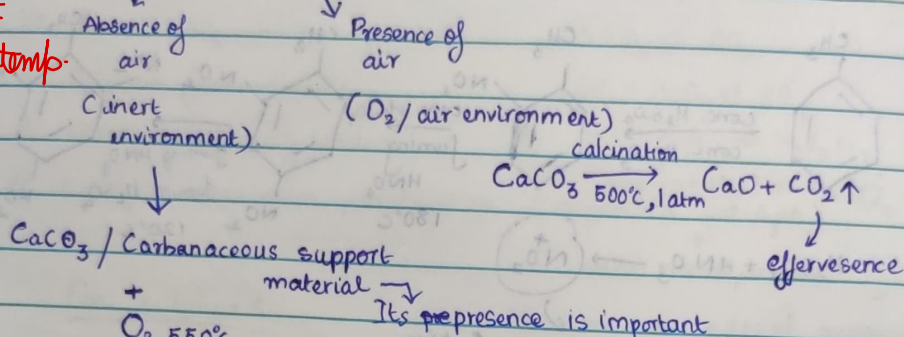
10/08/2022



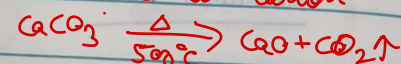
Support should have medium to high surface area ($>100 \text{ m}^2/\text{g}$)

Calcination reaction: $\rightarrow$ used in cement industry

heat treatment at certain temp. (below mp)



a) If our material is non carbonaceous doesn't contain carbon.

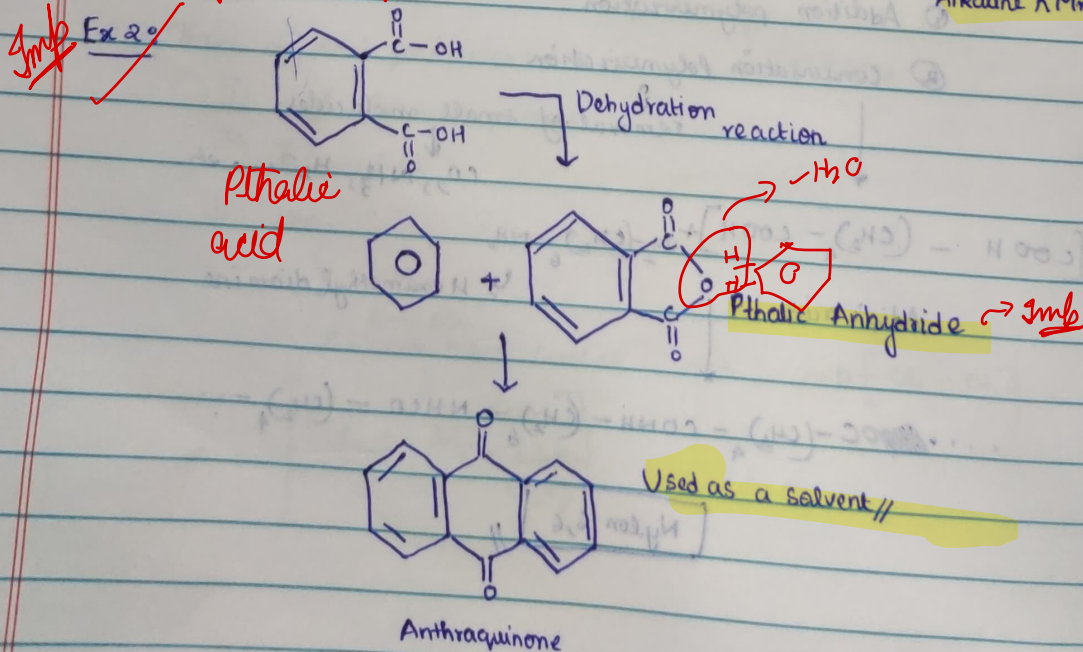
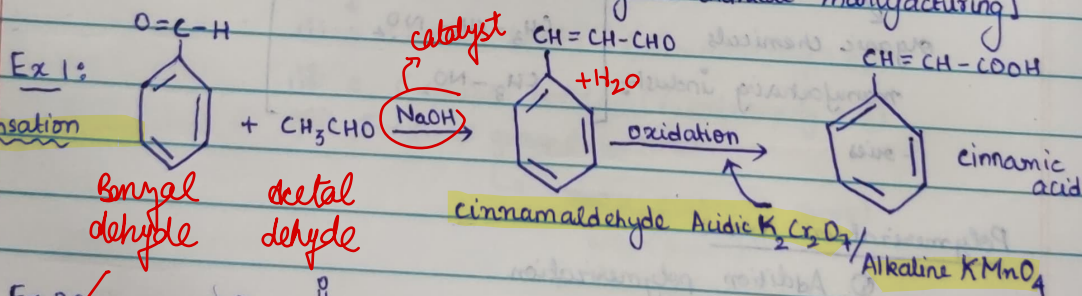


Complete oxidation: $\text{CO}_2 \uparrow$ (Carbon Dioxide)

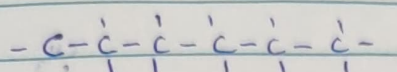
Partial Oxidation: $\text{CO} \uparrow$ (Carbon Monoxide)

b) If carbonaceous support material CaCO_3 / carbon $\xrightarrow[\text{1 atm}]{\text{500-550}^\circ\text{C, N}_2}$ CaO / carbon

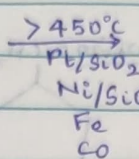
Condensation Reaction: [Used in organic chemical manufacturing]



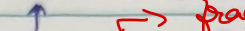
Catalytic cracking: (Used in the petroleum industry)



Paraffine compound



one is saturated



fragmentation

[P.N: These are hydrocarbons with three carbon atoms]

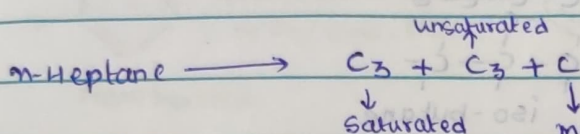
one is unsaturated

3 carbon atom product

catalytic cracking

If we use noble metals $\rightarrow$ less loading for higher conversion

If we replace with transition metals $\rightarrow$ more loading for higher conversion

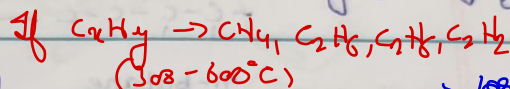


unsaturated

saturated

methane

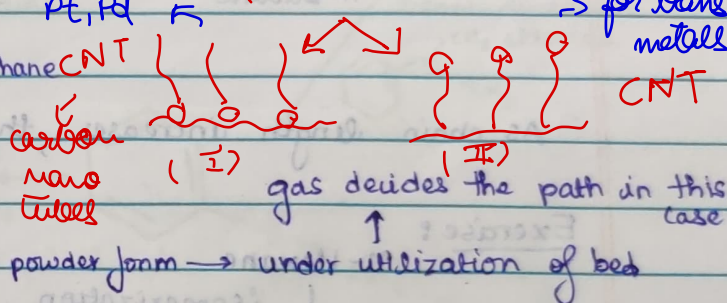
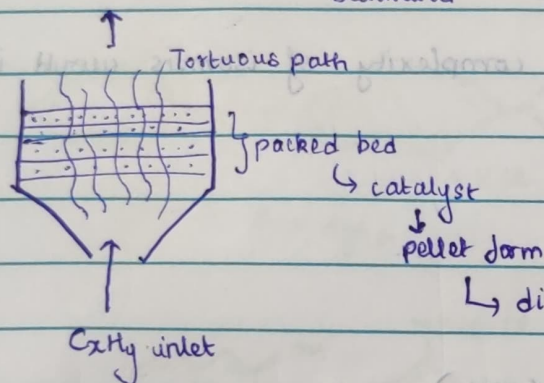
for Pt, Al



(300-600°C)

for transition metals

CNT



diameter should be roughly 3 mm

the pellets decides the gas path

(Detection of H_2 using Gas chromatography)

Carbon Nano-tubes (CNT)

2 Growth mechanisms: $\rightarrow$ Tip growth

Base growth

interaction with support

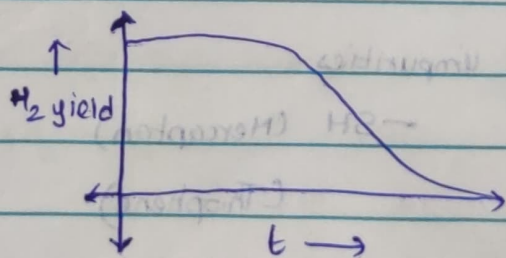
The metal is on the outer

surface of the support

interaction with active metal

they capture the tips of the CNT

Yield decreases :-



because carbon gets deposited, creating a barrier

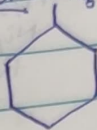
Catalytic Reforming: (Used in petroleum refineries)

actual
material

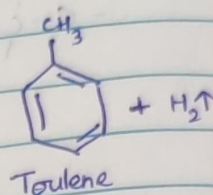
Paraffin dehydrocyclization: CH_3

n-Heptane

cyclization

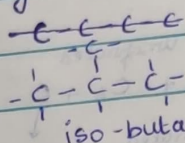
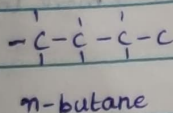


Dehydrogenation



Isomerization Reaction:

(no of C & H atoms remain same)



As chain length increases, the complexity of isomers would increase.

Exercise:

n-Hexane

↓ isomerization

?

Reactions related to petroleum refinery (cont.)

Without impurities (rxns)

Isomerization, Cracking, Cyclization, Dehydro Paraffin Dehydrocyclization

With impurities

↳ Feed contains (S, O, N) impurities

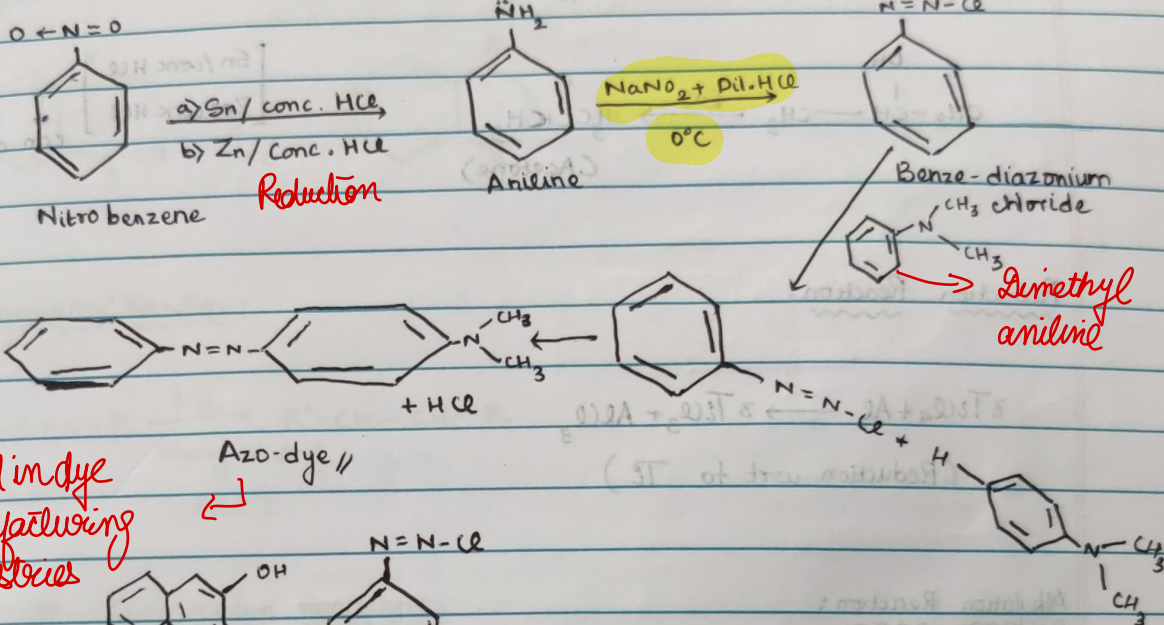
Feed (C, S, O, N) → Purification → Purified feed product

↳ containing impurities

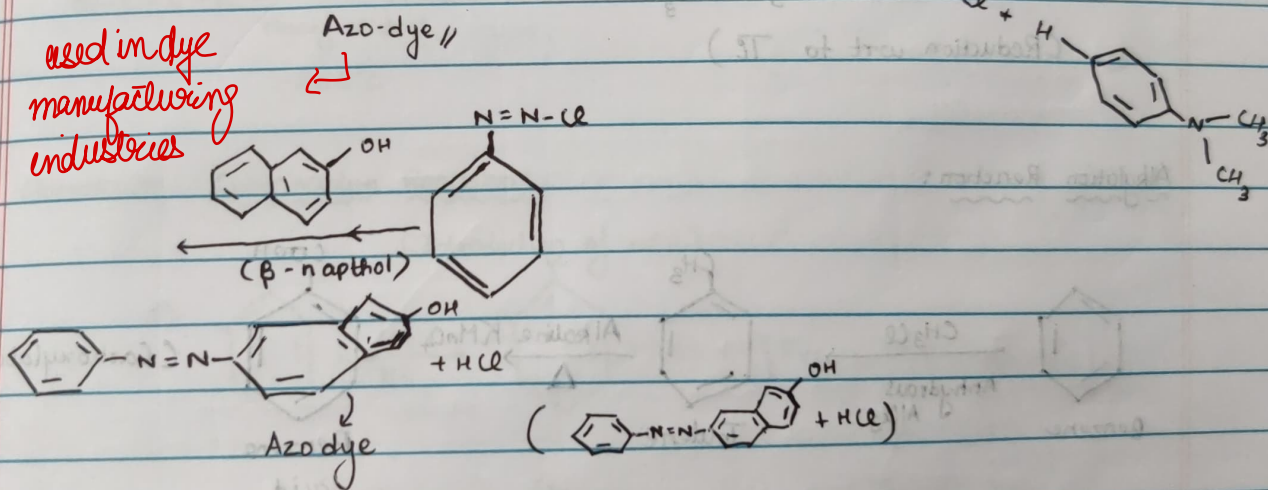
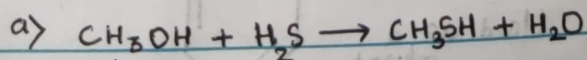
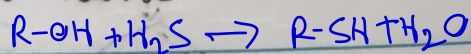
—SH (Mercaptan)

(Thiophene)

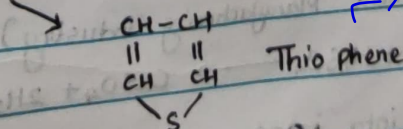
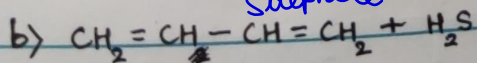
Feed $\xrightarrow[\text{High pressure}]{\text{H}_2}$ H_2S + Sulphur-free product

Diazotization reaction (Paint manufacturing industry)

used in dye manufacturing industries

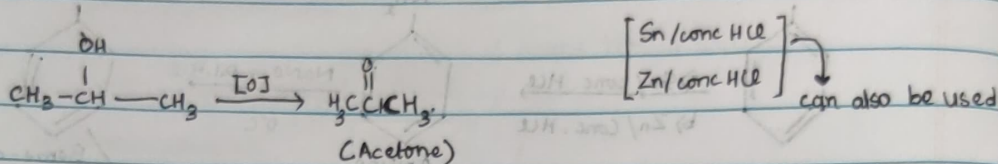
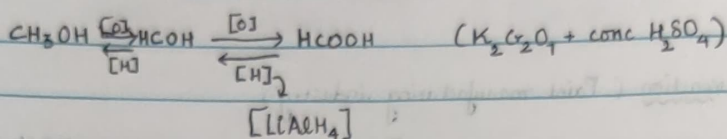
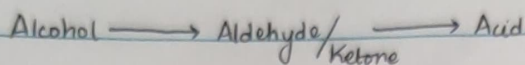
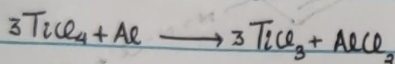
Thiomation Reaction : $\rightarrow$ Incorporation of $-\text{S}-\text{SH}$ groups

methanol $\downarrow$ hydrogen sulphide (methyl mercaptan)

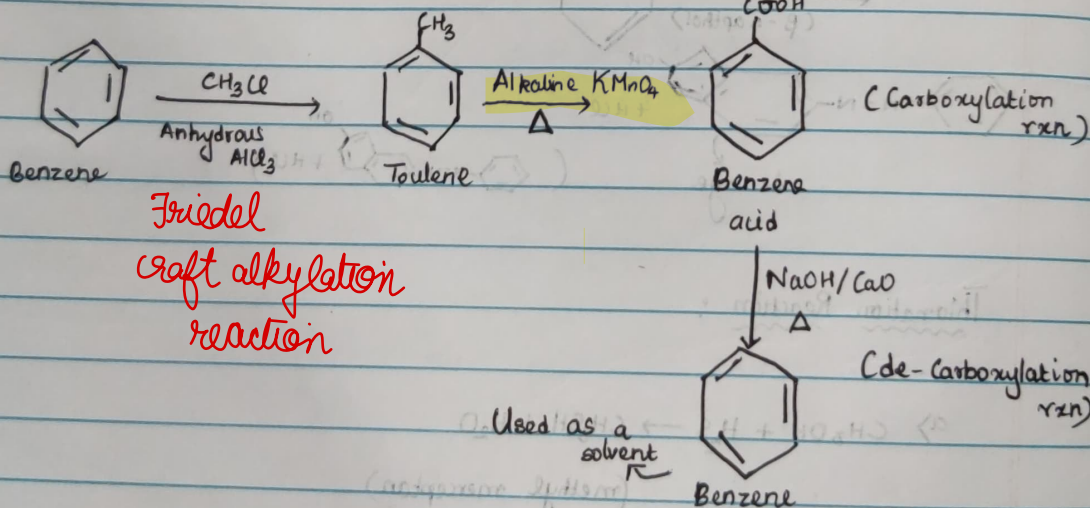


impurity in crude oil

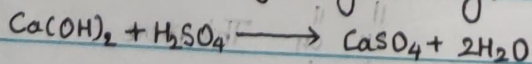
hydrodesulphurization $\rightarrow$ use of H to remove S containing impurities in petroleum industry

Oxidation Reaction:Reduction Reaction:

(Reduction wrt to Ti)

Alkylation Reaction:Double Decomposition Rxn

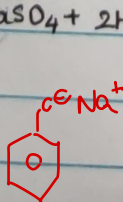
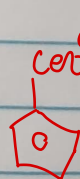
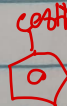
(Cement Manufacturing Industry)

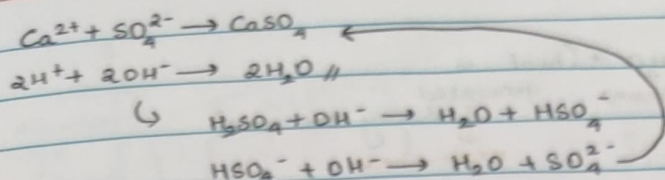


a) Break it into ions //

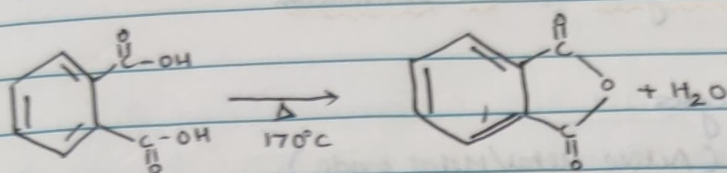
sodium salt of phenol

sodium phenoxide

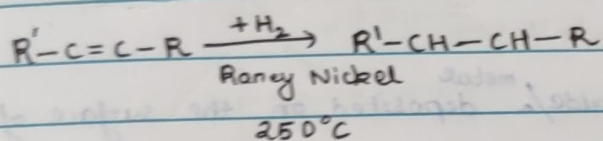
+ CO₂
high pressure+ H⁺



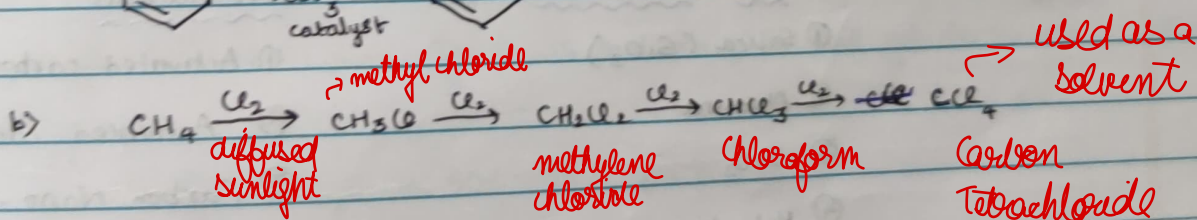
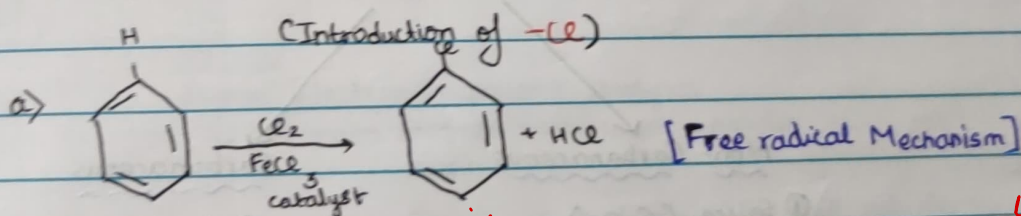
Dehydration Reaction:



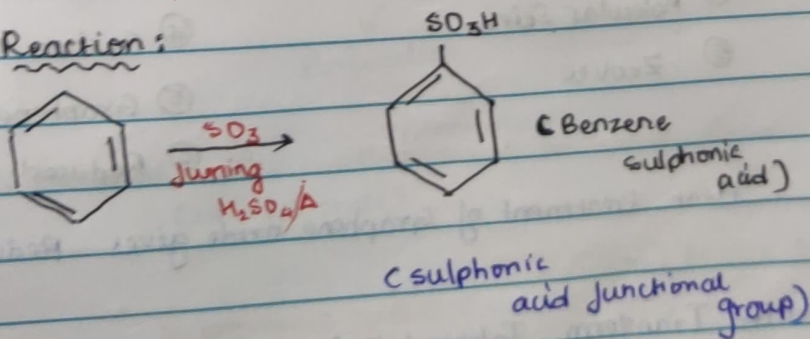
Hydrogenation Reaction:



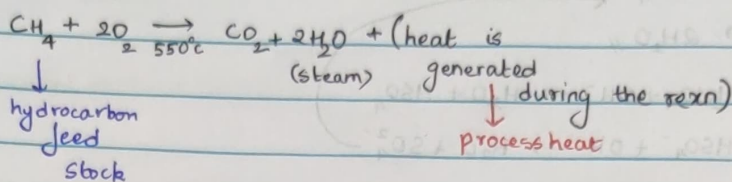
Chlorination Reaction:



Sulphonation Reaction:

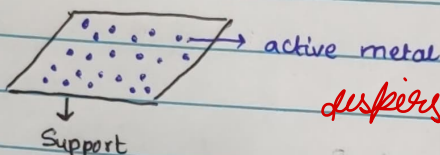


Combustion Reaction



Catalysis & Its Deactivation

- Focus on supported catalysts
- Unsupported catalysts (Active Metal/Metal oxide)



Dispersion: Fraction of metal oxide/metal deposited on the surface of the support

dispersion - fraction of active metals deposited on surface of support

Properties of support material

- medium to high surface area
- High melting pt
- support should be less active than active metal

Support Catalyst

→ high surface area facilitates dispersion of active material

Non-Carbonaceous

- Silica (SiO_2)
- Alumina
- ZrO_2 , TiO_2 also
- Molecular Sieve
- Zeolite

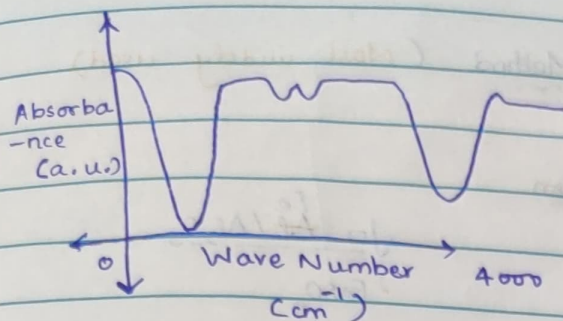
Carbonaceous

- Activated carbon → black carbon particles
- Activated carbon fibres
- Carbon Nano-tube → tubular shaped carbon particles
- Carbon Nano-fibres
- Graphene/Graphene oxide
- Reduced Graphene Oxide

Heat treatment of Graphene oxide gives Reduced Graphene Oxide.

FTIR (Fourier Transform Infrared Radiation)

↓
To identify the functional groups



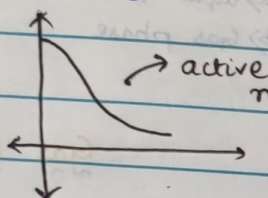
{ Range b/w 450 & 4000 cm^{-1} }

burn-off

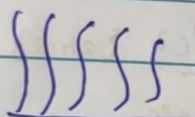
(At high temperatures)

$\text{C} \rightarrow \text{CO}_2 \uparrow$ (If we use carbonaceous support)

Therefore it is preferable to use non-carbonaceous support



All the carbonaceous support burns off.



(CNTs) //

SEM $\rightarrow$ Scanning Electron microscopic analysis
TEM $\rightarrow$ Transmission Electron microscopic analysis

The catalyst must be pure. If it is not pure, it will get deactivated relatively easily.

If catalyst is 90% - 95% pure

Purification

STEP-1

Support Material + H_2O (drop wise) $\rightarrow$ Paste
in a crucible (1g - lab scale)

Volatiles Impurities $\downarrow$ & entrapped gases
 CO_2 vapour

(4hrs) Muffle Furnace | presence of air
for 4 hours $550^\circ\text{C}/\Delta$

Remove crucible (after cooling)
Solid lumps
Paste in (lump form)

(High purity) Powder Support material

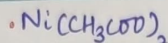
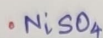
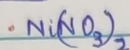
Ball Milling

Removal of volatile impurities & entrapped gas

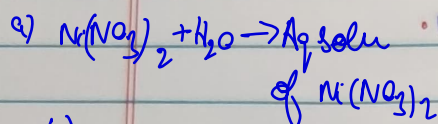
Incipient Wetness Impregnation Method (Most widely used)

① Precursor Salt solution $\rightarrow \text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$

for $\text{Ni}/\text{Al}_2\text{O}_3$

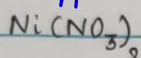


for $\text{Fe}/\text{Al}_2\text{O}_3$

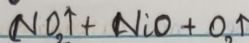


6) Basis: 1g of support material for $\text{Pt}/\text{Al}_2\text{O}_3$

c) $\text{Ni}(\text{NO}_3)_2$ - support



→ to change Ni^{+2} to Ni^0
(calcination (500°C) CO_2 environment)



Reduction



Ni -Support

↳ $\text{Ni}(\text{NO}_3)_2$ will get deposited onto support

Liquid Phase

a) NaBH_4

b) Hydrazine Mono-hydrate
 $\text{C}_2\text{H}_4 \cdot \text{H}_2\text{O}$

Gas Phase

a) H_2

↳ $\{550^\circ\text{C}\}$ $\{2\text{hrs}\}$

For $\text{Zn} \rightarrow 700^\circ\text{C}$

For $\text{Cu} \rightarrow 300^\circ\text{C}$

Tubular furnace

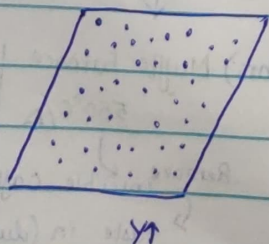
Composite Catalyst

Support

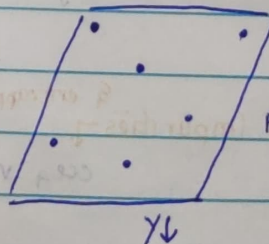
Active metal

↓
type of applications

Now our aim would be to disperse the active metals onto the support material



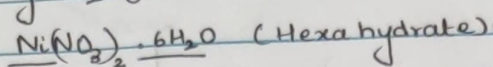
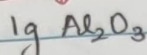
Metals are dispersed



Fewer active sites

(powder form)

lg support

 $\rightarrow$ aq. $\text{Ni(NO}_3)_2$ solutionThere will be no bond-formation with solid $\text{Ni(NO}_3)_2 + \text{Al}_2\text{O}_3 //$ (ii) Mixing of Al_2O_3 and aq. $\text{Ni(NO}_3)_2$ $\downarrow$ drop-wise $\text{Ni(NO}_3)_2 / \text{Al}_2\text{O}_3 \rightarrow$ thoroughly mixed

(calcination)

(paste) $\swarrow$ muffle furnace in presence of O_2
duration: 4hrs

Reduction

 $\xleftrightarrow[2\text{h}]{550^\circ\text{C}}$  $\downarrow$
dispersion
in Al_2O_3 $\rightarrow$ Either use gas phase (H_2)
orLiquid phase (NaBH_4)

High grade stain-less steel

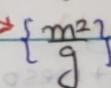
Inconel Reactor $\rightarrow$ for reduction $\downarrow$
Leak free $\rightarrow$ Hydrogen is explosiveComposite Catalyst:

Experiments to test physicochemical properties

 $\downarrow$
a) Surface area & pore size distribution (BET method)

b) Acidic & basic properties (TPD analysis)

c) Surface functional groups

 $\downarrow$
Temperature Programmed
Desorption Analysis

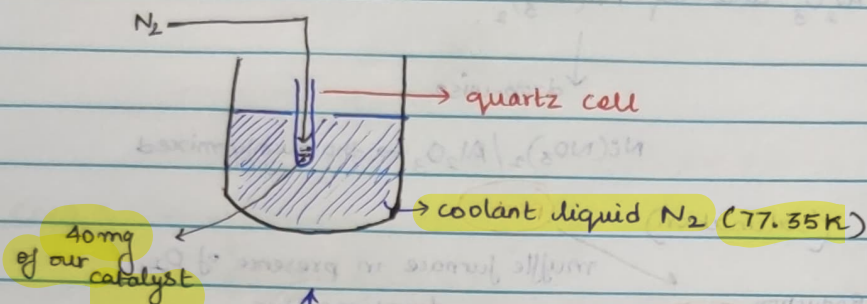
3 Different pores:

(i) Micro pores $< 2 \text{ nm}$ ex carbonaceous(ii) Meso pores $2 \text{ to } 50 \text{ nm}$ (iii) Macro pores $> 50 \text{ nm}$

BET:

- ① Surface area (m^2/g)
- ② Total pore volume (cc/g)
↓
 $\Sigma (\text{Micro} + \text{Meso} + \text{Macro})$ pores
- ③ Average pore diameter ($\text{\AA}/\text{nm}$)

These can be obtained through
BET
 Brunauer - Emmett - Teller

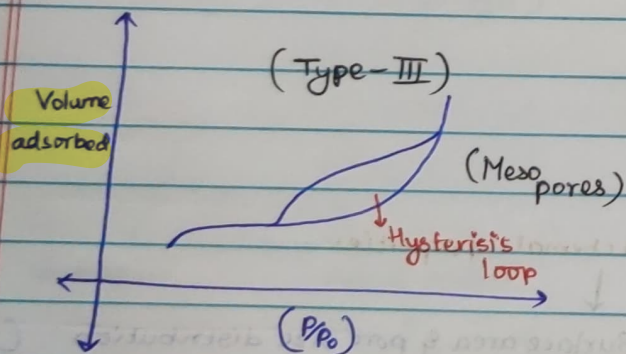


Volume
of N_2
adsorbed

(Type-I)

carbanaceous material

Microporous in nature

 $(P/P_0) \rightarrow 760 \text{ mmHg}$ N_2 - adsorption/desorption isotherms

For BET will plot from relative pressure of 0 to 0.3

because it is linear in this range

$$\text{Macro} = \text{TPV} - \underbrace{(\text{CM} + \text{M})}_{\text{micro + meso}}$$

Physico-chemical properties:

a) Surface area and pore size distribution: (S_{BET} & PSD)

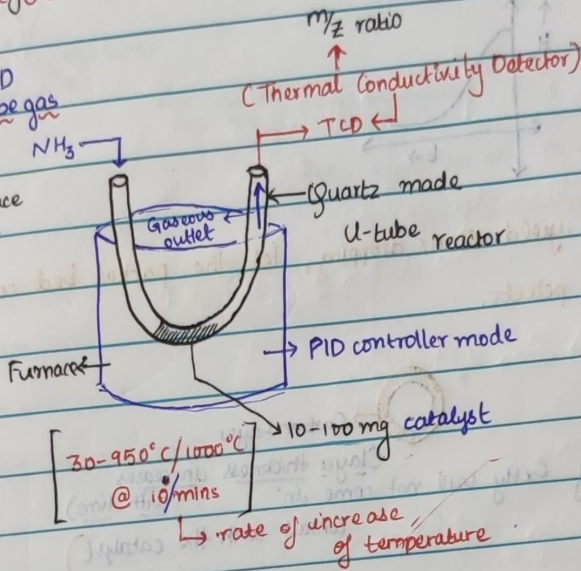
b) Surface properties $\rightarrow$ Acidic
 $\rightarrow$ Basic

c) FTIR (Determination of surface functional groups)

Temperature Programmed Desorption (TPD) analysis

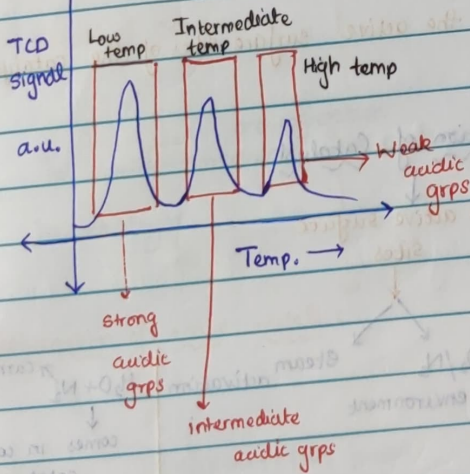
$\downarrow$
 NH_3 as a TPD probe gas

$\downarrow$
 Determine acidic surface sites



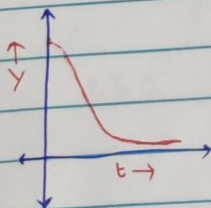
CO_2 as a TPD probe gas

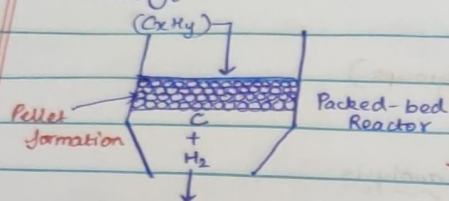
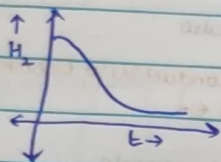
$\downarrow$
 Determine basic surface sites



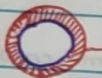
Catalyst Deactivation:

Catalyst Deactivation $\rightarrow$ coking $\rightarrow$ sintering $\rightarrow$ poisoning



a) Coking:Temperature of reactor: 450°C - 500°C 

Once yield starts dropping, let the packed bed cool, so as to remove the pellets.



Carbon Layer

(Incoming C_xH_y will not come in contact with the catalyst)
 Layer thickness increases with time

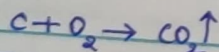
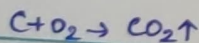
Coking $\rightarrow$ deposition of coke on the active surface sites of the catalyst pellets.

Regeneration of Catalyst

retain active surface sites

In O_2/N_2 environment

Steam activation

 O_2 (200 - 250°C) N_2 (500°C)

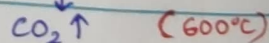
TGA

 30 - 1150°C

(Thermo gravimetric analysis)

 $\text{H}_2\text{O} + \text{N}_2$ carrier gas

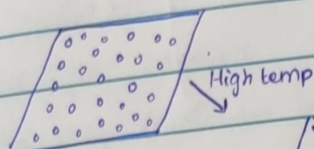
comes in contact with catalyst

b) Sintering:

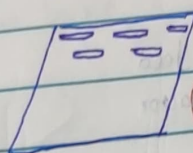
At high temperature ($> 600^\circ\text{C}$)

Narrowing of pores

Aggregation of the active metal



High temp



(pore diameter reduces)

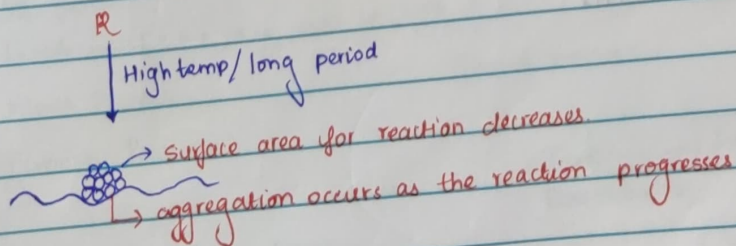
(C_6H_6 won't be able to diffuse through pore-mouths)
↓
unavailability of active-sites

7 steps for reaction on porous catalyst:

1. Diffusion from bulk
2. Diffusion to pore mouth
3. ~~Diffusion~~ Adsorption onto pore
4. Reaction
5. }
6. } Reverse of 3, 2 & 1
7. }

To avoid narrowing and aggregation
0.3-0.4 times lesser than melting point temperature must be employed.

Aggregation of active metals:



c) Poisoning:

P.T.O.

Feed ($\text{CS}_2, \text{O}_2, \text{N}_2, \text{Ne} \dots$)
(Impure)

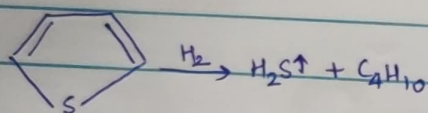
(Pre treated feed)
 $\downarrow$
heat

Purification of feeds:

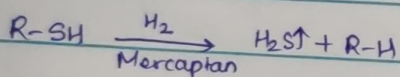
Feed $\rightarrow$ Pre-treater $\rightarrow$ Pure feed
Reactor

(Hydrodesulphurization)

Addition of H_2 at high P



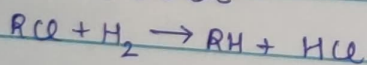
Thiophene



End uses:

- ① $\text{SO}_2, \text{SO}_3, \text{H}_2\text{SO}_4$ (-90%), $\text{CS}_2, \text{P}_2\text{S}_5$
- ② Gun powder ($\text{C} + \text{S} + \text{KNO}_3$)
- ③ Paints and platties containing S

Hydrodechlorination:



5 Manufacturing

• S is used to produce H_2S and H_2SO_4

• Properties of S: Atomic weight = 32.07

↓
Rhombic (112.8°C) Monoclinic (119°C) Allotropic forms

• Boiling point = 444.6°C

• Solubility (insoluble in H_2O , S is soluble in organic solvents $\xrightarrow{NH_3}$ NH_4OH)

• Different forms:

- Rock
- Molten
- Lump
- Fine powder

Consumption Pattern:

(80-90)% of S is consumed in the form of oxides

1. End uses:

- ① Fertilizer industries
- ② Inorganic chemical manufacturing industries
- ③ Organic chemical manufacturing industries
- ④ Paper & pulp industry
- ⑤ Polymer industry

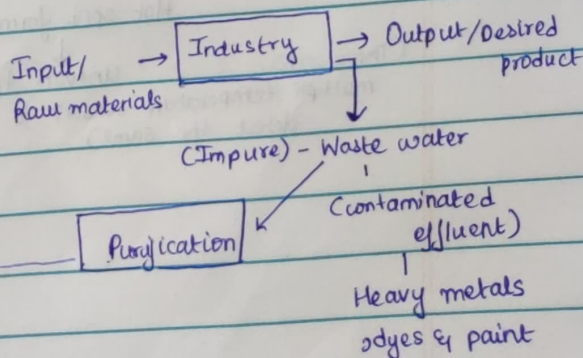
Synthesis of Sulphur:

* Frasch process → Only used in USA

* Klaus Process

* Finnish Process

General process:

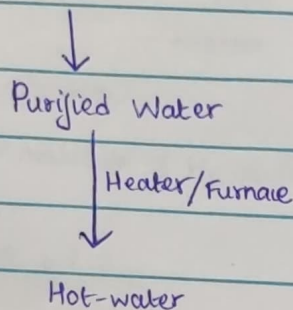
Raw materials in Frasch Process:

a) Sulphur bearing salt domes

b) Hot-water

Possible steps to obtain pure water, to recycle used water:

- Chlorination rxn
reaction
 - Fentan rxn → Fentan reagent ($\text{FeSO}_4 + \text{H}_2\text{O}_2$)
 - Adsorption
 - Advanced oxidation Process
↳ catalytic oxidation process
(High Pressure O_2 is passed through water)
- } only one of the methods will be used

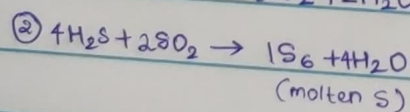
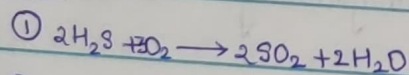


Klaus Process: (Oxidation & Reduction of H_2S)

Raw materials:

H_2S (Natural gas / Petroleum refining process)

Reaction:



{ 2 step - catalytic reactor }

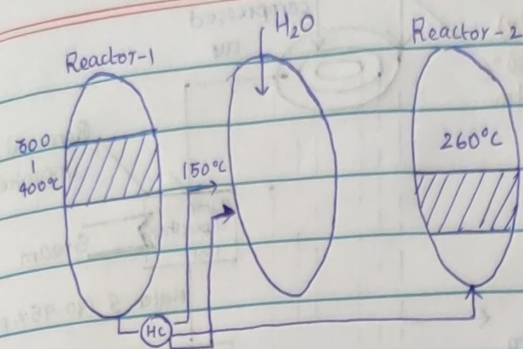
constructed with Stainless Steel

Reactor-1 → (300-400°C) [Bauxite catalyst, 70-80% conversion]

Reactor-2 → (250-300°C) [Al_2O_3 , 95-99% conversion]

Hot-spot formation:

↓
uneven temperature distribution
(There are multiple temperature sensors to detect the same)

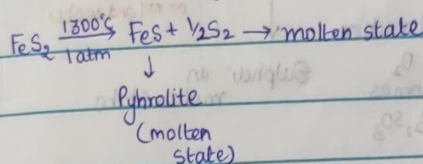


Finnish Process

Iron pyrites (FeS_2)

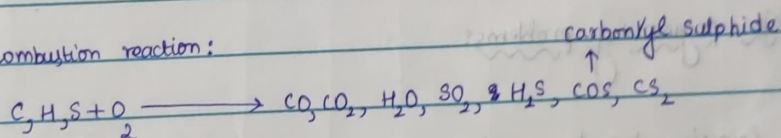
↓
40-45% sulphur

① Thermal dissociation of FeS_2 :

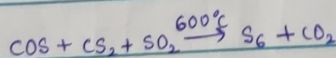


Endothermic rxn

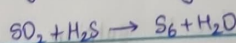
② General combustion reaction:



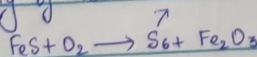
③ Recovery of sulphur from hot-stage: (600°C , Bauxite catalyst)



④ Recovery of sulphur from cold stage: (Klaus)



⑤ Roasting of FeS : Molten state



↓
steel manufacturing industries

12/09/2022:

Frasch Process

Only used in USA

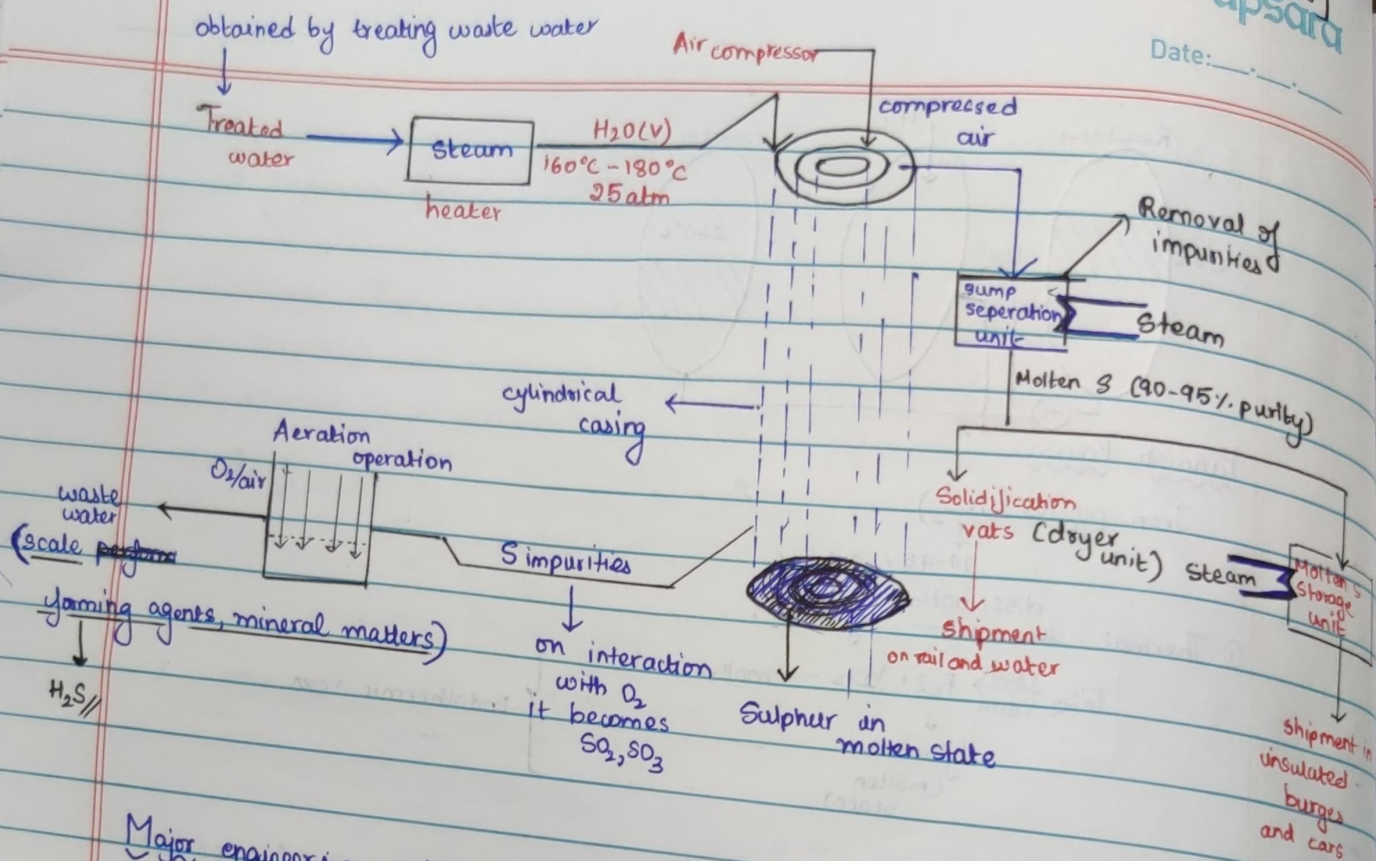
Basic raw-materials:

- a) Sulphur bearing salt domes
- b) steam

[Filtration is done to remove carbonaceous impurities / acidic impurities]

apsara

Date: _____



Major engineering problems:

- Corrosion due to sulphur
- Heat loss during the transport of molten S
- Suitable sources of waste water.

Finding

Klaus Process:

a) Oxidation

Major Problems:

- Corrosion due to sulphur